
Intelligence Engineering

Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | September 25, 2026 · *drifting stream*

LECTURE XI

The Human Side of Machine Learning

This lecture in one slide

XI The Human Side of Machine Learning

User Experience

Team Structure

Responsible AI

*why this lecture matters, and what
you can do after it*

1

User Experience

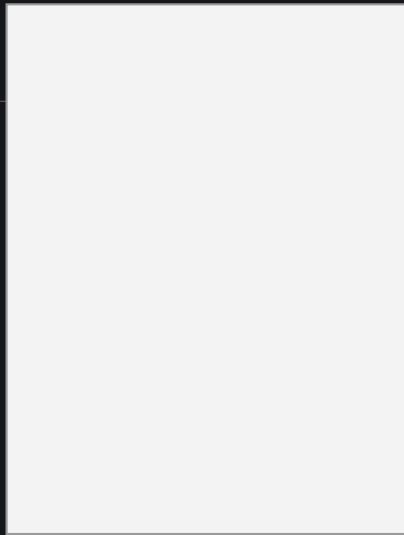
Ensuring User Experience Consistency

$P(y|x)$

[1]

Combatting “Mostly Correct” Predictions

Mostly correct predictions



$P(y | x)$

[1]

Smooth Failing

Definition: Smooth failing

2

Team Structure

Cross-functional Teams Collaboration

$P(y|x)$

[1]

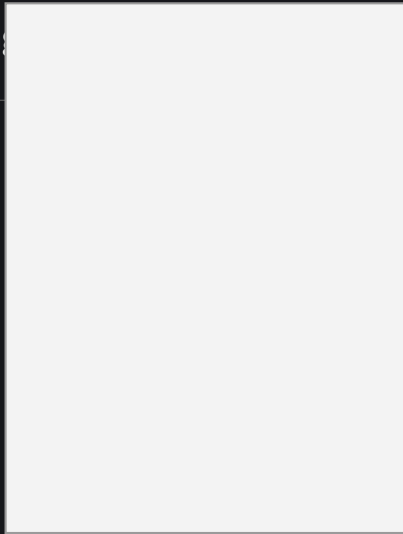
End-to-End Data Scientists

Approach 1

Approach 2

Approach 1: Have a separate team to manage

A separate team to manage production



Approach 2: Data scientists own the entire process

$P(y | x)$

[1]

3

Responsible AI

Irresponsible AI: Case Studies

Case study I

Case study II

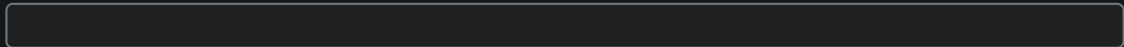
Case study I: Automated grader's biases

$P(y|x)$

[1]

Case study II: The danger of “anonymized” data

$P(y | x)$



[1]

A Framework for Responsible AI

$P(y|x)$

[1]

Discover sources for model biases

$P(y|x)$

Understand the limitations of the data-driven approach

$P(y | x)$

[1]

Understand the trade-offs between different desiderata

Desideratum	Trade-off

Act early

$P(y|x)$

Act early

[1]

Create model cards

Model card

$P(\hat{y} | x)$

[1]

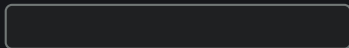
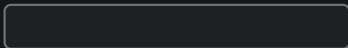
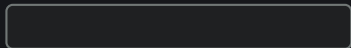
Establish processes for mitigating biases

$P(y|x)$

[1]

Stay up-to-date on responsible AI

$P(y|x)$



[1]

4

Summary

Summary

$P(y|x)$

[1]

References

- 1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.